

h MATH 1 FINAL COACHING

1. Given the current $P = I(20-3I)$, find the maximum current when $t = 0$ s?

- A. $10/3$
- B. $3/10$
- C. 10
- D. 3

2. Find the time at which the current is maximum if the current is given by $i(t) = t e^{(-0.5t)}$

- A. 1
- B. 2
- C. 4
- D. 0.53. $\iint_R (x+y) dA$ where
- E. the region R is: $0 \leq x \leq 2, 0 \leq y \leq 3$

Answer: 15

4. Find the length of arc of $y = x^2$ from $x = 0$ to $x = 1$:

The arc length of the curve $y = x^2$ from $x = 0$ to $x = 1$ is:

- A. $\int_0^1 \sqrt{1+x^2} dx$
- B. $\int_0^1 \sqrt{1+4x^2} dx$ ✓
- C. $\int_0^1 1+2x dx$
- D. $\int_0^1 x^2 dx$

5. Find the length of arc of $y = x^3$ from $x = 0$ to $x = 1$:

Find the length of arc of $y = x^3$ from $x = 0$ to $x = 1$:

- A. $\int_0^1 \sqrt{1+9x^4} dx$
- B. $\int_0^1 \sqrt{1+3x^2} dx$
- C. $\int_0^1 (1+9x^4) dx$
- D. $\int_0^1 \sqrt{1+6x} dx$

ANS: A

6. Find the length of arc of $y = x^n$

Find the length of arc of $y = x^n$:

- A. $\int \sqrt{1 + nx^{n-1}} dx$
- B. $\int \sqrt{1 + n^2 x^{2n-2}} dx$
- C. $\int (1 + n^2 x^{2n-2}) dx$
- D. $\int \sqrt{1 + n^2 x^{n-2}} dx$

~B;

ANS: B

7. Find the surface area generated when the curve $y=e^x$ from $x=0$ to $x=1$ is revolved about the x-axis

Find the surface area generated when the curve $y = e^x$ from $x = 0$ to $x = 1$ is revolved about the x-axis:

- A. $2\pi \int_0^1 e^x \sqrt{1 + e^{2x}} dx$
- B. $2\pi \int_0^1 e^x \sqrt{1 + e^x} dx$
- C. $\pi \int_0^1 e^{2x} \sqrt{1 + e^{2x}} dx$
- D. $2\pi \int_0^1 \sqrt{1 + e^{2x}} dx$

~A;

8. The best method to evaluate $\int x^2 e^x dx$ is:

- A. Substitution
- B. Partial Fractions
- C. Integration by Parts**
- D. Trigonometric Substitution

The best method to evaluate $\int x^2 e^x dx$ is:

- A. Substitution
- B. Partial Fractions
- C. Integration by Parts
- D. Trigonometric Substitution

9. The best method to evaluate $\int e^{(2x)} x dx$ is:

- A. Substitution
- B. Partial Fractions
- C. Integration by Parts (repeated)**
- D. Trigonometric Substitution

The best method to evaluate $\int x e^{2x} dx$ is:

- A. Substitution
- B. Partial Fractions
- C. Integration by Parts (repeated)
- D. Trigonometric Substitution

10. Best method to evaluate $\int 5x+1/x^2 + 4 dx$:

- A. Integration by Parts
- B. Partial Fractions
- C. Algebraic Substitution**
- D. Trigonometric Substitution

The best method to evaluate $\int \frac{5x+1}{x^2+4} dx$ is:

- A. Integration by Parts
- B. Partial Fractions
- C. Algebraic Substitution
- D. Trigonometric Substitution

$+ C$

11. Which of the following equations represents Faraday's Law for a time-varying magnetic field?

- A. $\nabla \cdot \mathbf{E} = \rho/\epsilon_0$
- B. $\nabla \times \mathbf{E} = -\partial \mathbf{B}/\partial t$**
- C. $\nabla \times \mathbf{B} = \mu_0 \mathbf{J}$
- D. $\nabla \cdot \mathbf{B} = 0$

12. Which of the following represents Gauss's Law for electricity?

- A. $\nabla \cdot \mathbf{E} = \rho/\epsilon_0$
- B. $\nabla \times \mathbf{E} = -\partial \mathbf{B}/\partial t$
- C. $\nabla \times \mathbf{B} = \mu_0 \mathbf{J}$
- D. $\nabla \cdot \mathbf{B} = 0$

13. What is the divergence of the magnetic field?

- A. $\nabla \cdot \mathbf{B} = \rho/\epsilon_0$
- B. $\nabla \cdot \mathbf{B} = 0$**
- C. $\nabla \times \mathbf{B} = \mu_0 \mathbf{J}$
- D. $\nabla \times \mathbf{E} = -\partial \mathbf{B}/\partial t$

14. What is the power series equation for arctan (x)?

What is the power series expansion of arctan(x) about x = 0?

- A. $\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{2n+1}$
- B. $\sum_{n=0}^{\infty} \frac{x^{2n+1}}{2n+1}$
- C. $\sum_{n=0}^{\infty} \frac{(-1)^n x^n}{n}$
- D. $\sum_{n=0}^{\infty} \frac{x^n}{n!}$

~A;

15. What is the Maclaurin series expansion of ln(1 + x)?

What is the Maclaurin series expansion of $\ln(1 + x)$?

A. $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{x^n}{n}$

B. $\sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1}$

C. $\sum_{n=0}^{\infty} \frac{x^n}{n!}$

D. $\sum_{n=1}^{\infty} \frac{x^n}{n}$

~A;

16. A machine depreciates at a rate of 9% per year. How long will it take for the machine's value to decrease to 75% of its original value?

Ans: 15.4 years

17. A population follows the logistic growth model but is subject to a constant harvesting rate H . The population dynamics are described by: $dP/dt = rP(1 - P/K) - H$, where r is the intrinsic growth rate and K is the carrying capacity. Under what condition will the population eventually go extinct?

A. $H < rP/K$

B. $H = rP/K$

C. $H > rP/K$

D. $H = rK$

18. Using the bisection method, determine the first approximation of the root of the equation $f(x) = x^2 - 3 = 0$ given the initial interval $[1, 2]$.

A. 1.50

B. 1.75

C. 1.25

D. 2.00

19. Using the bisection method, determine the second approximation of the root of the equation $f(x) = x^2 - 3 = 0$ given the initial interval $[1, 2]$.

A. 1.50

B. 1.75

C. 1.25

D. 1.625

20. Let the scalar function be $f(x,y,z) = xy + yz + xz$. Find the directional derivative of f at the point $P(1,1,1)$ in the direction of the vector $\mathbf{v} = \langle 1,0,1 \rangle$.



Problem: Directional Derivative

Let the scalar function be

$$f(x, y, z) = xy + yz + xz$$

Find the **directional derivative** of f at the point

$$P(1, 1, 1)$$

in the direction of the vector

$$\mathbf{v} = \langle 1, 0, 1 \rangle.$$

Ans: $2\sqrt{2}$

21. A machine costs \$50,000 and depreciates continuously at a rate of 10% per year. Its value $V(t)$ after years is modeled by $d/dt(V) = -0.10V$. How long will it take for the machine's value to drop to \$20,000?

Ans: 9.16 yrs.

22. Which differential equation governs the current $i(t)$ in a series RL circuit driven by a DC voltage source V_s ?

Ans:

$$L \frac{di}{dt} + Ri = V_s$$

23. A series RL circuit has $R = 4\Omega$ and $L = 2H$. A 1-V DC source is applied at $t = 0$. Determine the expression for the current $i(t)$ in the circuit for $t > 0$.

Ans: $\frac{1}{4} (1 - e^{-2t})$

24. The discrete cross-correlation between two sequences $x[n]$ and $h[n]$ is defined as:

- A. $r[n] = \sum x[k] h[n - k]$
- B. $r[n] = \sum x[k] h[k - n]$**
- C. $r[n] = \sum x[n] h[k]$
- D. $r[n] = \sum x[k - n] h[k]$

25. In convolution, the standard discrete-time formula between two sequences $x[n]$ and $h[n]$ is defined as:

- A. $y[n] = \sum x[k] h[k - n]$
- B. $y[n] = \sum x[k] h[n - k]$**
- C. $y[n] = \sum x[n] h[k]$
- D. $y[n] = \sum x[k - n] h[k]$

26. If $\text{cov}(X, Y) = -2$, $\text{var}(X) = 4$, and $\text{var}(Y) = 9$, what is the correlation coefficient $\text{cor}(X, Y)$?

Ans. $-\frac{1}{3}$

27. If the σ_x is 2, the σ_y is 3, and the $\text{cor}(x, y)$ is 0.5, find the covariance between X and Y .

- A. 6
- B. 3**
- C. 12
- D. 24

28. If $\text{cor}(X, Y) = 0.2$, $\text{var}(X) = 16$, and $\text{var}(Y) = 25$, what is $\text{cov}(X, Y)$?

- A. 2
- B. 4**
- C. 6
- D. 8

29. In a planar PCB, the scalar potential is $V(x, y) = x^2 + xy + y^2$. Find the directional derivative at $(1, 2)$ in the direction of vector $\mathbf{v} = (2, 1)$.

Ans: 13/√5

30. A manufacturer claims that **5% of its products are defective**. A random sample of **100 products** is inspected and **2 are found to be defective**. At the **5% significance level**, test the claim that the defect percentage has **changed**.

- A. The defect percentage increased
- B. The defect percentage decreased
- C. The defect percentage changed
- D. The defect percentage remained unchanged

A manufacturer claims that 5% of its products are defective ($p_0 = 0.05$). A random sample of 100 products is inspected and 2 are found to be defective. At the 5% significance level, test the claim that the defect percentage has changed.

- A. The defect percentage increased
- B. The defect percentage decreased
- C. The defect percentage changed
- D. The defect percentage remained unchanged

~D;

$$z = \frac{\bar{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$$

31. A 2^2 factorial experiment is conducted to study the effects of factors A and B on a response variable. The factors are tested at two levels coded as -1 and +1. The observed responses are shown below:

A	B	Responses
-1	-1	12
+1	-1	20
-1	+1	16

+1	+1	24
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Determine the sum of squares due to factor B (SSB).

- A. 4
- B. 8
- C. 16**
- D. 32

32. A 2^2 factorial experiment is conducted to study the effects of factors A and B on a response variable. The factors are tested at two levels coded as -1 and $+1$. The observed responses are shown below:

A	B	Responses
-1	-1	20
+1	-1	28
-1	+1	18
+1	+1	26

Determine the sum of squares due to factor A (SSA).

- A. 16
- B. 32
- C. 64**
- D. 128

33. A 2^2 factorial experiment is conducted to study the effects of factors A and B on a response variable. The factors are tested at two levels coded as -1 and $+1$. The observed responses are shown below:

A	B	Responses
-1	-1	12
+1	-1	20
-1	+1	16
+1	+1	30

Determine the sum of squares of the interaction (SSAB).

- A. 1

- B. 2
- C. 6
- D. 8

34. A spring-mass-damper system has mass $m = 1$ kg, damping coefficient $c = 4$ Ns/m, and spring constant $k = 13$ N/m. The system is subjected to a step force $f(t) = 10u(t)$. If the system starts from rest, determine the displacement equation $x(t)$.

A spring-mass-damper system has mass $m = 1$ kg, damping coefficient $c = 4$ N·s/m, and spring constant $k = 13$ N/m. The system is subjected to a step force $f(t) = 10u(t)$. If the system starts from rest, determine the displacement equation $x(t)$.

- A. $x(t) = \frac{10}{13} \left[1 - e^{-2t} \left(\cos 3t + \frac{2}{3} \sin 3t \right) \right]$
- B. $x(t) = \frac{10}{13} e^{-2t} \left(\cos 3t + \frac{2}{3} \sin 3t \right)$
- C. $x(t) = \frac{10}{13} \left[1 - e^{-2t} \left(\cos 3t - \frac{2}{3} \sin 3t \right) \right]$
- D. $x(t) = \frac{10}{13} \left[1 + e^{-2t} \left(\cos 3t + \frac{2}{3} \sin 3t \right) \right]$

~A;

35. Determine the **radius of convergence** of the power series:

Determine the **radius of convergence** of the power series

$$\sum_{n=1}^{\infty} \frac{(x+2)^n}{n 3^n}$$

- A. 1
- B. 2
- C. 3
- D. ∞

~C;

36. What is the Interval of Convergence for the power series

49 Find the interval of convergence of the power series of:

$$\sum_{n=1}^{\infty} \frac{(x-2)^n}{n \cdot 3^n}$$

[-1,5)

I forgot the solution

Answer: [-1,5)

37. Given the discrete sequence $x(n) = (1, 2, 1)$, determine the autocorrelation at lag 1.

- A. 2
- B. 3
- C. 4**
- D. 5

38. Determine the cross-correlation of the sequences $x = [1, 2]$ and $y = [2, 1]$ at lag 0

- A. 2
- B. 3
- C. 4**
- D. 5

39. The velocity error constant K_v of a control system with open loop transfer function $G(s)$ is defined as:

ANS: B

The velocity error constant K_v of a control system with open-loop transfer function $G(s)$ is defined as:

- A. $K_v = \lim_{s \rightarrow 0} G(s)$
- B. $K_v = \lim_{s \rightarrow 0} sG(s)$
- C. $K_v = \lim_{s \rightarrow 0} \frac{G(s)}{s}$
- D. $K_v = \lim_{s \rightarrow 0} s^2 G(s)$

40. A periodic signal $f(t)$ is represented by its Fourier coefficients as follows: $a_0 = 2$, $a_1 = 4$, $b_1 = 4$

Ans.: 16

41. Given the transfer function, determine the steady-state output of the system.

$$G(s) = \frac{5}{2s + 5}$$

Ans. 1

42. A solid is formed by erecting **square cross-sections (cubes)** perpendicular to the x-axis, where the **side of each square is given by $f(x)$** . Which of the following represents the **volume of the solid**?

ANS: B

A solid is formed by erecting **square cross-sections (cubes)** perpendicular to the x-axis, where the **side of each square is given by $f(x)$** . Which of the following represents the **volume of the solid**?

A. $V = \int f(x) dx$

B. $V = \int [f(x)]^2 dx$

C. $V = \frac{\sqrt{3}}{4} \int [f(x)]^2 dx$

D. $V = \pi \int [f(x)]^2 dx$

43. A solid is formed by erecting **equilateral triangular cross-sections perpendicular to the x-axis**, where the side of each triangle is given by $f(x)$. Which of the following represents the **volume of the solid**?

A solid is formed by erecting **equilateral triangular cross-sections perpendicular to the x-axis**, where the side of each triangle is given by $f(x)$. Which of the following represents the **volume of the solid**?

A. $V = \int [f(x)]^2 dx$

B. $V = \frac{\sqrt{3}}{4} \int [f(x)]^2 dx$

C. $V = \frac{1}{2} \int [f(x)]^2 dx$

D. $V = \pi \int [f(x)]^2 dx$

~B;

44. The **x-coordinate of the centroid ($\bar{x}$)** of a region bounded by the curve $y=f(x)$, the x-axis, and the lines $x=a$ and $x=b$ is given by which of the following formulas?

The **x-coordinate of the centroid** ($\bar{x}$) of a region bounded by the curve $y = f(x)$, the x-axis, and the lines $x = a$ and $x = b$ is given by which of the following formulas?

A. $\bar{x} = \frac{\int_a^b x f(x) dx}{\int_a^b f(x) dx}$

B. $\bar{x} = \frac{\int_a^b f(x) dx}{\int_a^b x f(x) dx}$

C. $\bar{x} = \frac{\int_a^b x dx}{\int_a^b f(x) dx}$

D. $\bar{x} = \int_a^b x f(x) dx$

~A;

45. Determine the Fourier transform of signal

A rectangular pulse is defined as

$$x(t) = \begin{cases} 5, & |t| \leq 2 \text{ ms} \\ 0, & \text{otherwise} \end{cases}$$

Determine the **Fourier transform of the signal**.

A. $10 \operatorname{sinc}(2t)$

B. $20 \operatorname{sinc}(4t)$

C. $5 \operatorname{sinc}(2t)$

D. $10 \operatorname{sinc}(4t)$

~B;

46. What is the integral of $\cos(ax) dx$?

- A. $(1/a) \sin(ax) + C$**
- B. $-(1/a) \cos(ax) + C$
- C. $(1/a) \tan(ax) + C$
- D. $-(1/a) \sin(ax) + C$

47. What is the integral of $\sin(ax) dx$?

- A. $(1/a) \sin(ax) + C$
- B. $-(1/a) \cos(ax) + C$**
- C. $(1/a) \tan(ax) + C$
- D. $-(1/a) \sin(ax) + C$

48. What is the integral of $\tan(ax) dx$?

- A. $-(1/a) \ln|\cos(ax)| + C$**
- B. $(1/a) \ln|\sin(ax)| + C$
- C. $(1/a) \ln|\tan(ax)| + C$
- D. $-(1/a) \ln|\cot(ax)| + C$

49. A system has the transfer function $G(s) = s / s(s + 2)$. If the input to the system is a unit step, determine the steady-state output.

- A. 0**
- B. 1
- C. $\frac{1}{2}$
- D. 2

50. A unity feedback control system has a position error constant $K_p = 4$. Determine the steady-state error for a unit step input.

- A. $\frac{1}{4}$
- B. $\frac{1}{5}$**
- C. $\frac{1}{6}$
- D. $\frac{1}{8}$

51. A unity feedback system has $K_p = \infty$. Determine the steady-state error for a ramp input.

- A. 0**
- B. 1
- C. ∞
- D. $1/2$

52. Determine the energy of the discrete sequence $x[n] = [1, 2, 1]$ using Parseval's theorem.

- A. 4**

- B. 5
- C. 6**
- D. 8

53. What is the **Laplace transform** of the time-shifted function $g(t) = f(t - a)u(t - a)$?

- A. $F(s)$
- B. $F(s)e^{-as}$**
- C. $sF(s)e^{-as}$
- D. $e^{as}F(s)$

54. Given the sequence $| -1 - n/n^2 |$ the sequence is said to be:

- A. Diverging
- B. Converging**
- C. Oscillating
- D. Undefined

55. Which of the following series is **absolutely convergent**?

- A. $\sum 1/n$
- B. $\sum 1/n^2$**
- C. $\sum (-1)^n / n$
- D. $\sum 1$

56. Which of the following represents the divergence of an electrostatic field?

- A. $\nabla \cdot \mathbf{E} = \rho/\epsilon_0$**
- B. $\nabla \times \mathbf{E} = -\partial \mathbf{B} / \partial t$
- C. $\nabla \times \mathbf{B} = \mu_0 \mathbf{J}$
- D. $\nabla \cdot \mathbf{B} = 0$

57. In an underdamped RLC circuit, the natural frequency is given by which of the following

In an **underdamped RLC circuit**, the natural frequency is given by which of the following?

- A. $\frac{1}{\sqrt{LC}}$
- B. $\sqrt{\frac{1}{LC} - \left(\frac{R}{2L}\right)^2}$
- C. $\frac{R}{2L}$
- D. $\sqrt{\frac{R}{LC}}$

58. If the damping ratio satisfies $0 < \zeta < 1$, the system is said to be:

- A. Undamped
- B. Underdamped**
- C. Critically damped
- D. Overdamped

59. If the damping ratio corresponds to the fastest response without oscillation, the system is said to be:

- A. Undamped
- B. Underdamped
- C. Overdamped
- D. Critically damped**

60. An engineer installed a new pump with a flow rate of $60\text{m}^3/\text{day}$ the flow rate decreases 15% per year assuming the decay follows continuous exponential model how many year will it pump to decrease to $20\text{m}^3/\text{day}$

- A. 5.5 years
- B. 7.3 years**
- C. 9.2 years
- D. 12.0 years

61. An radioactive substance has an intial mass of 200g it continuously decay 12% per year how long will it take for the substance to decrease to 50g

- A. 9.3 years
- B. 10.8 years
- C. 11.55 years**
- D. 13.2 years

62. Given in the table below, find the F ratio

	SS	df	
Treatment	48	8	?
Error	32	4	?
Total			

63. In an experiment the $MST = 50$ and $MSE = 10$, the F ratio is?

- A. 5**
- B. 10
- C. 15
- D. 25

64. In ANOVA, the F-ratio is computed as which of the following?

- A. $(SS_{\text{treatment}} / DF_{\text{treatment}}) / (SS_{\text{error}} / DF_{\text{error}})$
- B. $(SS_{\text{error}} / DF_{\text{error}}) / (SS_{\text{treatment}} / DF_{\text{treatment}})$
- C. $(SS_{\text{total}} / DF_{\text{total}}) / (SS_{\text{error}} / DF_{\text{error}})$
- D. $(SS_{\text{treatment}} / DF_{\text{error}}) / (SS_{\text{error}} / DF_{\text{treatment}})$

65. Find the Laplace transform of $e^{(-2t)} \sin(3t)$.

- A. $3 / ((s + 2)^2 + 9)$
- B. $(s + 2) / ((s + 2)^2 + 9)$
- C. $3 / ((s - 2)^2 + 9)$
- D. $(s - 2) / ((s - 2)^2 + 9)$

66.

Question: Two RL circuits share a mutual resistor R_m :

$$\begin{cases} L \frac{di_1}{dt} + (R + R_m)i_1 - R_mi_2 = V_1(t) \\ L \frac{di_2}{dt} + (R + R_m)i_2 - R_mi_1 = V_2(t) \end{cases}$$

Which method is most appropriate to solve this system?

- A. Laplace transformation
- B. Separation of variables
- C. Euler's method
- D. Integration by parts

Answer: A

67.

Question: A mass-spring-damper system is subjected to a harmonic forcing function $F \cos(\omega t)$:

$$m\ddot{x} + c\dot{x} + kx = F \cos(\omega t)$$

Which statement is true about the **steady-state oscillating frequency**?

- A. The steady-state oscillating frequency equals the natural frequency $\sqrt{k/m}$
- B. The steady-state oscillating frequency equals the forcing frequency ω
- C. The steady-state oscillating frequency depends on the damping c
- D. The steady-state oscillating frequency is zero

Answer: B

68.

Question: Temperature of an object cools in an environment with temperature $T_a(t) = 20 + 5 \sin(\omega t)$.

Which method is appropriate?

- A. Laplace transformation
- B. Separation of variables
- C. Integrating factor for a non-homogeneous linear DE
- D. Characteristic equation method

Answer: C

69.

Question: For a mass-spring-damper system under $F \cos(\omega t)$, which method can be used to find the steady-state amplitude?

- A. Laplace transform
- B. Undetermined coefficients
- C. Separation of variables
- D. Euler's method

Answer: B

70.

Question: A capacitor with capacitance C and resistance R is discharging from an initial voltage V_0 . What is the equation for the capacitor voltage $V(t)$ as a function of time?

- A. $V(t) = V_0(1 - e^{-t/RC})$
- B. $V(t) = V_0 e^{-t/RC}$
- C. $V(t) = V_0 e^{t/RC}$
- D. $V(t) = V_0(1 + e^{-t/RC})$

Answer: B

71.

41. Find the Fourier transform of the rectangular pulse:
 $x(t) = \begin{cases} 2, & 0 \leq t \leq 1 \text{ ms} \\ 0, & \text{otherwise} \end{cases}$

Nakalimutan ko solution pero eto answer

$$X(f) = 2T \text{sinc}(fT) e^{-j\pi fT}$$

72.

42. Using the vertex method, find the minimum value of the quadratic function:
 $-2x^2 + 8x + 3$

$$\begin{aligned} &= -\frac{b}{2a} \\ &= -\frac{8}{2(-2)} \\ &= 2 \end{aligned}$$

73. Rectangular pulse, using parsevals theorem find the total energy

45. Using the Parseval theorem, find the total energy of

$$x(t) = \begin{cases} 1, & |t| \leq 1 \\ 0, & |t| \geq 1 \end{cases}$$

$$\begin{aligned} E &= \int_{-\infty}^{\infty} |x(t)|^2 dt \\ &= \int_{-1}^1 1^2 dt \\ &= 2 \end{aligned}$$

74.

Question: What is the first-order differential equation of a series RL circuit that governs the current $i(t)$?

A. $L \frac{di}{dt} + Ri = V(t)$

B. $L \frac{di}{dt} + \frac{R}{L}i = V(t)$

C. $R \frac{di}{dt} + Li = V(t)$

D. $L \frac{d^2i}{dt^2} + Ri = \frac{dV(t)}{dt}$

Answer: A

Explanation: For series RL: $V(t) = L di/dt + Ri$. This is a **first-order linear DE**.

75.

Question: What is the first-order differential equation of a series RC circuit that governs the current $i(t)$?

A. $RC \frac{di}{dt} + i = V(t)$

B. $R \frac{di}{dt} + \frac{1}{C}i = \frac{dV(t)}{dt}$

C. $R \frac{di}{dt} + Ci = V(t)$

D. $R \frac{di}{dt} + \frac{1}{C} \frac{di}{dt} = V(t)$

Answer: B

76.

Question: What is the second-order differential equation of a series RLC circuit that governs the current $i(t)$?

A. $L \frac{d^2 i}{dt^2} + Ri + \frac{1}{C} i = V(t)$

B. $L \frac{di}{dt} + Ri + \frac{1}{C} \frac{di}{dt} = V(t)$

C. $L \frac{d^2 i}{dt^2} + R \frac{di}{dt} + \frac{1}{C} i = \frac{dV(t)}{dt}$

D. $L \frac{d^2 i}{dt^2} + R \frac{di}{dt} + Ci = V(t)$

Answer: C

77.

Question: What is the first-order differential equation of a series RC circuit in terms of the capacitor voltage $V_c(t)$, given resistance R and applied voltage $V(t)$?

A. $RC \frac{dV_c(t)}{dt} + V_c(t) = V(t)$

B. $R \frac{dV_c(t)}{dt} + V_c(t) = CV(t)$

C. $RC \frac{dV(t)}{dt} + V(t) = V_c(t)$

D. $R \frac{dV_c(t)}{dt} + CV_c(t) = V(t)$

Answer: A

78. A solid of revolution is formed by revolving the region bounded by $y = x^2$ about the y-axis. Which method is most appropriate to use?

- A. Washer Method
- B. Disk Method
- C. Shell Method**
- D. Cylindrical Method

79. The region bounded by $y = x^2$, $x = 1$ and $x = 4$ is revolved about the x-axis. Which method is most appropriate to use to find the volume?

- A. Shell Method
- B. Disk Method
- C. Washer Method**

D. Cylindrical Method

80.

Find the volume of the solid generated when the region bounded by $y = e^x$, $x = 0$, and $x = 1$ is revolved about the x-axis.

A. $V = \pi \int_0^1 e^{2x} dx$

B. $V = \pi \int_0^1 e^x dx$

C. $V = 2\pi \int_0^1 xe^x dx$

D. $V = \pi \int_0^1 xe^{2x} dx$

~A;